

Application of Artificial Intelligence and Big Data Technology in Digital Marketing

Fang Gao
Wuhan Business University,
Wuhan, Hubei, China
12027904@qq.com

Lan Zhang
Suizhou Vocational & Technical
College, Suizhou, Hubei, China
175426147@qq.com

ABSTRACT

At present, intelligent technologies such as artificial intelligence and big data are in full swing, and various application scenarios are gradually landing. The application of big data and artificial intelligence technologies to digital marketing has become a top priority for the development of the industry, and has been It has become a consensus and is becoming more and more popular. However, there are still many problems in the application process, the most common of which is that marketing is not accurate enough. Under this background, this article uses a hotel chain company as an example to study the precision of digital marketing, which makes the marketing effect significantly improved. The introduction of artificial intelligence and big data makes data no longer just a simple entry. These intelligent technologies fully tap the potential behind these data. These potentials have important values and significance for social development.

CCS CONCEPTS

•Applied computing •Operations research •Computer-aided manufacturing

KEYWORDS

Big data, Artificial intelligence, Precision, Digital marketing

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1 Introduction

With the rapid development of high and new technology, technology has become more and more integrated into and affect our lives. At present, information technology has been widely used in all walks of life, and has been fully applied in the marketing industry, especially in the digital marketing industry. application.

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The advent of the 5G era is rapidly compressing the distance of time and space, and mankind is stepping towards the vision of a global village step by step. The development of artificial intelligence has greatly improved the operational efficiency of each industry and subdivided fields. Especially in recent years, the research and application of artificial intelligence in customer service, marketing and other fields have gradually deepened, which has brought opportunities to this industry and market. And challenges. In such an era of rapid technological development, the pace of social operation, the ability of humans to process information, and the impact of science and technology on social progress are accelerating, and the rate of increase is unprecedented. This article introduces artificial intelligence and big data technology, and then analyzes the application of artificial intelligence and big data technology in digital marketing.

2 Artificial intelligence and big data technology

2.1 Artificial intelligence

Artificial intelligence technology, which is a brand new technical science, is mainly used for simulating and expanding people, including a series of related theories, methods, technologies and application systems. Artificial intelligence technology makes relevant recommendations and analysis by analyzing people's thinking, behavior, and actions. One of the core technologies of artificial intelligence is to train computers to learn human behavior and thinking methods, improve their learning ability and stored new technologies and knowledge through continuous training, and then improve the stored knowledge through computer language to enrich their skills. The computer can use its permanent memory to store a large amount of past information and continuously optimize its applications to improve performance standards.

2.2 Big data technology

Big data is an information asset, which has a new processing mode to have stronger analytical power, pertinence and optimization capabilities. It cannot be collected, processed and managed by traditional conventional applications. It is a kind of Diversified and constantly updated assets. According to the view in "The Age of Big Data" written by Victor Mayer-Schonberg and Kenneth Coker, big data uses all data for analysis and processing, rather than traditional random analysis or sample survey Methods. IBM proposes that big data has five characteristics: Value, Velocity, Variety, Volume, and Veracity. Big data technology is to process the meaning contained in these data on the basis of a large amount of data information. By combining its stored historical data with

real-time updated data for analysis and processing, the new model needed is further developed.

3 Features of intelligent digital marketing

3.1 Digital marketing

Digital marketing is a marketing method that uses digital techniques such as network technology, computer technology, multimedia technology and interactive technology to achieve marketing purposes. The goal of digital marketing is to maximize the use of computer technology to efficiently develop the market and tap consumer needs. Digital marketing utilizes digital multimedia methods such as Weibo, short video, WeChat, and email to achieve precise marketing to specific target users in the database. Digital marketing also has a series of advantages, such as low marketing costs, customization, high timeliness, and high user relevance. This digital communication channel promotes the practical activities of products and services, allowing merchants and consumers to communicate better. The details are shown in Figure 1.



Figure 1: Marketing system model

3.2 Consumer demand

Peter Drucker is a well-known master of management. He proposes that the ultimate goal of marketing is to gain insight and knowledge of users. The development of artificial intelligence technology and big data technology enables enterprises to understand users more fully, improve the quality of service to users, and at the same time can continuously improve the image of the enterprise itself in the minds of users. By recording the behavior of each user, using intelligent technology to "portrait" each user. Based on the user's behavior data, combined with artificial intelligence, big data and other technologies, the potential behavior activities behind these behavior data are tapped to make the marketers more targeted. Using artificial intelligence and big data technology, analyze and predict the user's future behavior trajectory, continuously enrich the user's "portrait" and resonate with the user, which can not only deepen the user's awareness of the enterprise, but also achieve the effect of improving marketing efficiency. Figure 2 shows.

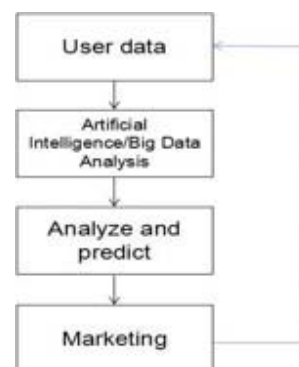


Figure 2: Consumer demand training model

3.3 Network traffic boosts digital marketing.

Currently, short videos are prevalent, and an important feature of these videos is entertainment. These videos first attracted attention through entertainment, and then slowly joined the marketing while updating the video. For example, we are familiar with "Douyin" and "Quick Hand", etc. In short video applications, you can often see the marketing of various products. Generally, the platform conducts promotion and marketing through analysis of personal browsing records and browsing data of users across the network, combined with personal preferences and the size of video traffic. In terms of online drama digital marketing, by continuously updating the repertoire on the Internet, it cultivates a large number of fans, protects the large flow of the repertoire, and at the same time continuously cultivates the demand of fans for products around the online drama.

4 Intelligent application of digital marketing

This article takes a hotel chain company as an example, and uses a clustering model to analyze intelligent digital marketing. The model is mainly composed of the following four algorithms.

The first is the K-means clustering algorithm. Clustering is to group a group of physical or abstract items into similar items. These groups are both meaningful and useful, or both. The basic idea of the algorithm is to find k Clusters, so that the records in each cluster are similar to each other and different from the records in other clusters. The second is the decision tree. The decision tree is a more intuitive and persuasive data algorithm. It can quickly develop and calculate data. This article selects a database-based decision tree, that is, chi-square automatic interactive detection (CHAID) decision tree. Next is the Pareto ratio, which states that nearly 80% of the effect comes from 20% of the reasons. This ratio is often used to express a large number of facts. This is a common rule of

thumb in business, such as the distribution of entrepreneurs and business managers. Explain that 80% of the profit comes from 20% of customers. Finally, the RFM model is widely used to evaluate the lifetime value of customers, and the clustering algorithm of the RFM model can provide the most powerful basis for research by constructing accurate objective variables.

A hotel company adopts the decision framework shown in Figure 3. The decision model mainly includes three parts, namely the data preparation part: preparing and analyzing the quality of customers to determine their goals; customer classification part:

using RFM model, clustering algorithm, decision tree algorithm, Pareto ratio to analyze each dimension Customer information to classify target customers; and finally make decisions based on the market.



Figure 3: Decision framework model

The data was first collected and 24173 customers were considered as the target customers of this case study. The results of data analysis using clustering algorithm are shown in Table 1. Then use the CHAID decision tree to determine the customers whose activities have higher order numbers, and the Pareto ratio to subdivide customer value, plus Response Models (Response Models) for payment prediction.

Table 1: Clustering algorithm analysis results

		Attributes		
		Cumulative number/Order amount	Cumulative number/Order amount	D value/week
Category 1	total	2028	2028	2028
	average value	102	19	3
	Maximum	61	27	24
	Minimum value	260	1	0
Category 2	total	82	82	82
	average value	382	20	1
	Maximum	897	27	18
	Minimum value	262	4	0
Category 3	total	2	2	2
	average value	382	20	1
	Maximum	897	27	18
	Minimum value	262	4	0
Category 4	total	22061	22061	22061
	average value	16	7	8
	Maximum	64	27	26
	Minimum value	1	1	0
overall	total	24173	24173	24173
	average value	24	8	8
	Maximum	1669	27	26
	Minimum value	1	1	0

With relatively mature intelligent data development technology, H Hotel Group uses CDP as a data platform to collect data in all directions and multiple dimensions, thereby discovering three types of potential customers with high value. These three types of

customers include: high-value commercial and travel customers 2. The customer groups and customers who intend to go out frequently on holidays. Then, based on statistical data, different marketing strategies were developed for different groups of people, the services required by customers were formulated, and the satisfaction of customer experience was improved. Perform programmatic advertising targeted activation, EDM information promotion, SMS information delivery, etc.

5 Conclusion

The era of intelligence has arrived, and artificial intelligence and big data technologies are leading the transformation of digital marketing to intelligent digital marketing. First, artificial intelligence and big data technology can make full use of existing marketing data and can be more convenient and efficient; second, artificial intelligence and big data technology can provide support for more accurate, targeted and creative digital marketing strategies.

When introducing artificial intelligence and big data technology into digital marketing, you also need to keep thinking. Although these intelligent technologies can greatly improve the efficiency, accuracy and diversification of digital marketing, but want to completely replace traditional labor with intelligent technology. The author believes that it is unlikely at least in the short term. Marketing is mainly about digging and researching user needs, while maintaining sufficient communication and communication, and this interactive behavior of communication and communication is difficult for a simple machine to completely replace manual. Therefore, it will take time for intelligent marketing such as artificial intelligence and big data to promote digital marketing.

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